



PROJECT PRESENTATION (PROJ-CS781) [2023-2027]

**PHISHGUARD: AN EXPLAINABLE MACHINE LEARNING FRAMEWORK FOR REAL-TIME
PHISHING URL DETECTION USING FEATURE ENGINEERING AND ENSEMBLE LEARNING**

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PRESENTATION OUTLINE

- Introduction
- Problem Statement
- Objective
- Methodology (System Architecture & Detection Workflow)
- Expected Result
- Project Milestone (Gantt Chart)
- Conclusion
- Limitations
- Future Work & References
- Q & A



INTRODUCTION

- **Phishing** is a cyberattack in which attackers create fake websites that imitate legitimate services to steal sensitive information such as passwords, banking credentials, OTPs, and personal data.
- **Unlike malware**, phishing relies primarily on social engineering and user trust, making it difficult to detect through content scanning alone.
- **PhishGuard** is proposed as an intelligent phishing URL detection framework that uses Feature Engineering, Machine Learning, and Ensemble Learning to identify suspicious URL patterns.
- **Instead of relying only on known malicious URLs**, the system extracts meaningful characteristics from a URL — its length, special characters, domain structure, subdomains, and IP-address usage.
- **An Explainable AI layer (SHAP)** is integrated so that every prediction is accompanied by a clear justification, rather than a plain phishing/legitimate label.



PROBLEM STATEMENT

Problem: Existing phishing detection systems struggle to identify newly generated, previously unseen, or deliberately obfuscated malicious URLs in real time.

Existing Challenges:

- **Zero-Day / New Phishing URLs:** Newly created malicious URLs do not exist in blacklists or reputation databases.
- **Evolving Attack Patterns:** Attackers continuously modify URL structures and use obfuscation to evade static rules.
- **High False Positives / False Negatives:** Distinguishing legitimate from malicious URLs accurately remains a major challenge.
- **Lack of Explainability:** Most systems only provide a prediction or warning without explaining why the URL is considered malicious.
- **Manual Analysis is Not Scalable:** Manual inspection is time-consuming, dependent on human expertise, and not scalable.



OBJECTIVE

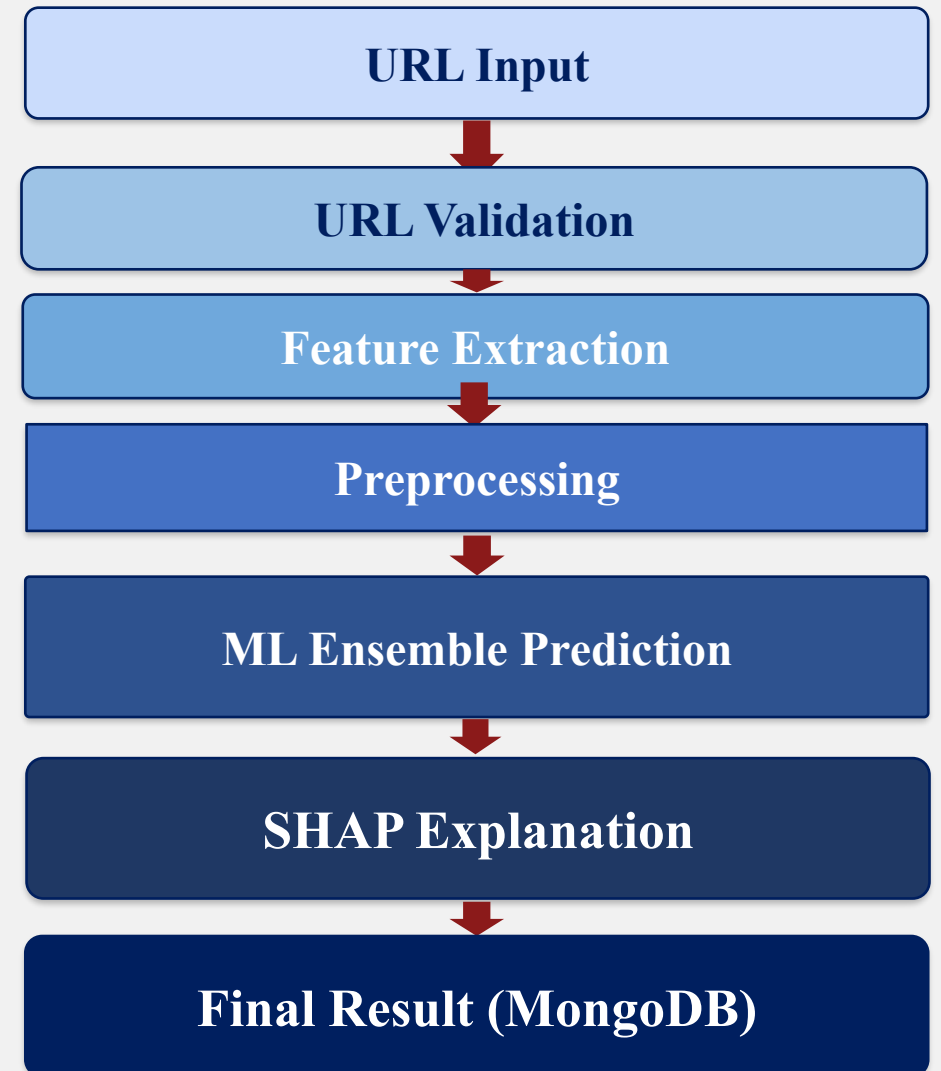
- **Feature Extraction:** Extract meaningful structural, lexical, and domain-related characteristics from URLs.
- **Feature Engineering:** Transform raw URL information into numerical features suitable for machine learning.
- **Phishing Classification:** Train machine learning models to accurately distinguish between phishing and legitimate URLs.
- **Ensemble Learning:** Combine predictions from multiple classifiers to improve robustness, accuracy, and reliability.
- **Explainable AI:** Use SHAP to identify and visualize the features that contribute to each prediction.
- **Real-Time Detection:** Provide an automated system that delivers instant phishing detection results through a web interface.



METHODOLOGY

- 1 URL Input & Validation**
The system provides a web interface where users submit a URL; the backend validates the input.
- 2 Feature Extraction & Engineering**
30+ lexical, domain-based, and structural features are extracted from the submitted URL.
- 3 ML Classification & Ensemble Learning**
Random Forest, XGBoost, SVM, LightGBM and CatBoost are combined through ensemble voting.
- 4 Explainable Prediction (SHAP)**
The system identifies which features contributed most to the classification.
- 5 Result Storage**
Prediction, confidence score, and explanation are stored in MongoDB for scan history.

PhishGuard Detection Pipeline





EXPECTED RESULT

- **High Detection Accuracy:** The ensemble model (Random Forest + XGBoost + SVM + LightGBM + CatBoost) is expected to achieve close to 95% accuracy in classifying phishing vs legitimate URLs.
- **Balanced Precision & Recall:** Target precision and recall above 90%, minimizing both false alarms and missed detections.
- **Transparent Explanations:** SHAP-based explanations are expected to clearly identify which URL features (length, IP usage, subdomains, etc.) drive each prediction.
- **Real-Time Performance:** End-to-end prediction is expected within a few hundred milliseconds, suitable for real-time API-based URL scanning.

[illegible]



CONCLUSION

- **Accurate & Explainable Detection:** PhishGuard combines feature engineering, ensemble machine learning, and SHAP-based explainability to detect phishing URLs with both accuracy and transparency.
- **Beyond Black-Box Predictions:** By explaining why a URL is flagged, PhishGuard builds user trust and supports faster security decision-making.
- **Practical, Deployable System:** The system is designed as an end-to-end web application (React + Django REST Framework + MongoDB) suitable for real-time use.



LIMITATIONS

- **Dataset Dependency:** Detection quality depends on the diversity and size of the labelled phishing/legitimate URL dataset used for training.
- **URL-Only Analysis:** The current system analyses URL-based features only, and does not inspect full webpage content or DNS/WHOIS data in real time.
- **Adversarial Evasion:** Attackers who deliberately craft URLs to mimic legitimate feature patterns may still evade detection.
- **Explainability Overhead:** Computing SHAP values adds processing overhead, which may affect very high-throughput deployments.



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Q & A



**Thank you for your attention.
We welcome your questions and
feedback.**



THANK YOU
FOR
LISTENING !!!